GROUP RESEARCH PROJECT

Reflections and Project Proposal Report

Student Name: Thaspeeha Vahithu
Student ID : 32146925
University : University of West London -RAK
Module Name: Group Research Project

TASK 1: (Individual Reflections Report)

Contents

1. Introduction 19

2. Evidence 19-14

2.1 Meetings Conducted 19

2.2 Evidence of Shared Documents 9

2.3 Evidence of each group member's contribution 14

3. Challenges Faced 14

4. Innovation and Attention To Detail 15-17

21.**1 Blog** **15**

4.2 Socials Page 15

4.3 Feedback Form 16

4.4 E-Gift Card 16

4.5 Sliding Gallery 16

4.6 Customer Reviews 17

4.7 Newsletter 17

4.8 Currency Converter 17

1. Introduction

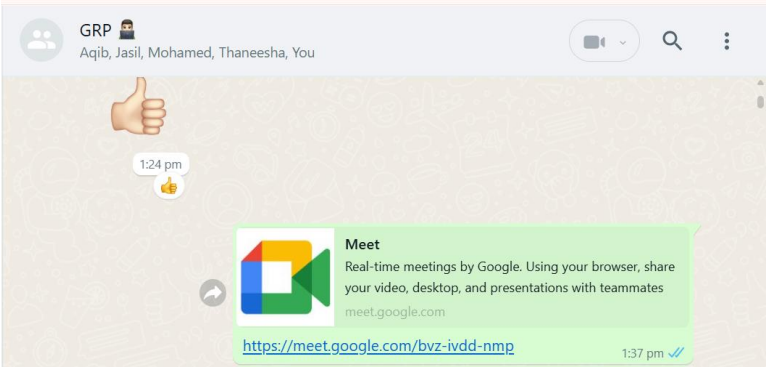
Our five-member team developed a user-friendly e-commerce website for Gear Up & Go. To keep the development process for the website organized and efficient, we assigned specific roles to each team member based on their strengths and expertise. This approach helped us work together seamlessly and stay on track to deliver our website on time.

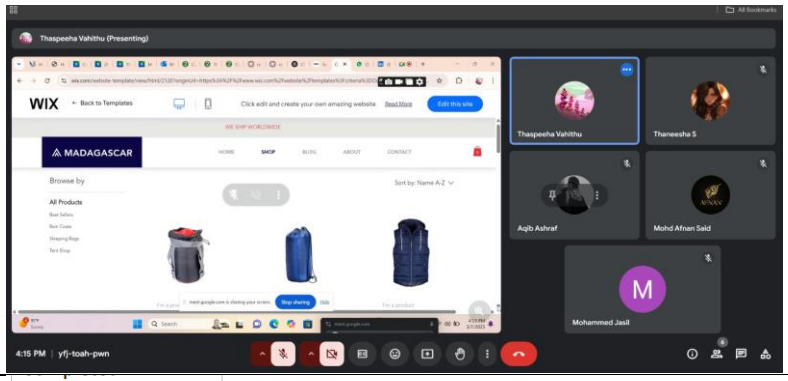
This report outlines and highlights our teamwork, how we each contributed to the project, and the ways we overcame challenges to create a quality e-commerce website. This report shows how we worked as a team, including evidence of our meetings, document sharing, and individual contributions and challenges were addressed to deliver a quality e-commerce platform tailored to the target audience and to achieve eco-friendly goals.

2.Evidence

Meetings Conducted (Screenshots)

<table>
<tr><th>Date</th><th>Mode</th><th>Minutes of Meeting</th></tr>
<tr><td>16/02/25 - 25/02/25</td><td>Offline Meeting</td><td>
<table>
<tr><td>Stage 1</td><td>PROJECT KICKOFF</td></tr>
<tr><td>Mode</td><td>In person at university</td></tr>
<tr><td>Status of attendence</td><td>All present</td></tr>
<tr><td>Date</td><td>Feb 16 – feb 25</td></tr>
<tr><td>Minutes</td><td>1. Decided on an adventure and outdoor gear e-commerce website as the project focus.
2. Finalized the website name: Gear Up & Go.
3. Confirmed 10 product categories, each containing 20 products.
4. Developed a task checklist to outline initial actions.</td></tr>
<tr><td>Challenges</td><td>Scheduling conflicts arose due to competing priorities.</td></tr>
<tr><td>Solutions</td><td>Conducted in-person meetings at university to ensure alignment.</td></tr>
<tr><td>Action items completed</td><td>Researched e-commerce platforms; finalized categories.</td></tr>
</table>
</td></tr>
<tr><td>26/02/25 -5/03/25</td><td>Online Meeting</td><td>
<table>
<tr><td>Stage 2</td><td>RESOURCE PLANNING</td></tr>
<tr><td>Mode</td><td>Online via Google meet</td></tr>
<tr><td>Status of attendence</td><td>All present</td></tr>
<tr><td>Date</td><td>Feb 26 – march 5</td></tr>
<tr><td>Minutes</td><td>1. Compiled high-resolution images for two categories per member (40 images each).
2. Selected Wix Studio as the development platform.
3. Designed a logo to establish brand identity.
4. Defined website structure and layout
5. Assigned roles and role specific tasks</td></tr>
<tr><td>Challenges</td><td>High-resolution images posed risks to website performance</td></tr>
<tr><td>Solutions</td><td>Implemented image compression to optimize load times.</td></tr>
<tr><td>Action Items Completed</td><td>Submitted images; prepared page layouts.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr>
<td></td>
<td></td>
<td>


(Google Meeting Link shared through WhatsApp)


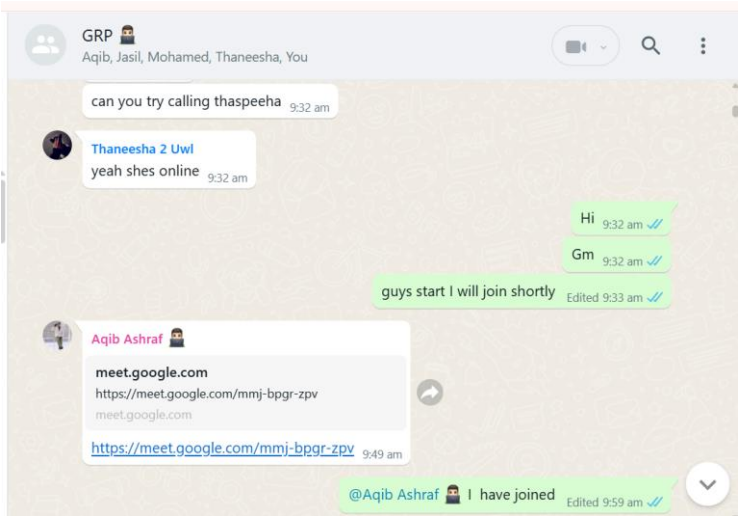
</td>
</tr>
<tr>
<td>06/03/25 - 09/03/25</td>
<td>Offline Meeting</td>
<td>
<table>
<tr><td>Stage 3</td><td>WEBSITE FRAMEWORK</td></tr>
<tr><td>Mode</td><td>In person at university</td></tr>
<tr><td>Status of attendence</td><td>All present</td></tr>
<tr><td>Date</td><td>March 6 – March 9</td></tr>
<tr><td>Minutes</td><td>1. Initiated website development on Wix Studio
2. Established pricing for all products
3. Produced product descriptions for all items.
4. Set clear deadlines to maintain momentum.</td></tr>
<tr><td>Challenges</td><td>Compiling accurate product specifications proved time-intensive.</td></tr>
<tr><td>Solutions</td><td>Consolidated data from reliable sources to ensure precision.</td></tr>
<tr><td>Action Items completd</td><td>Completed descriptions; distributed website draft.</td></tr>
</table>
</td>
</tr>
<tr>
<td></td>
<td></td>
<td></td>
</tr>
</table>

10/03/25 - 15/03/25	Online Meeting	

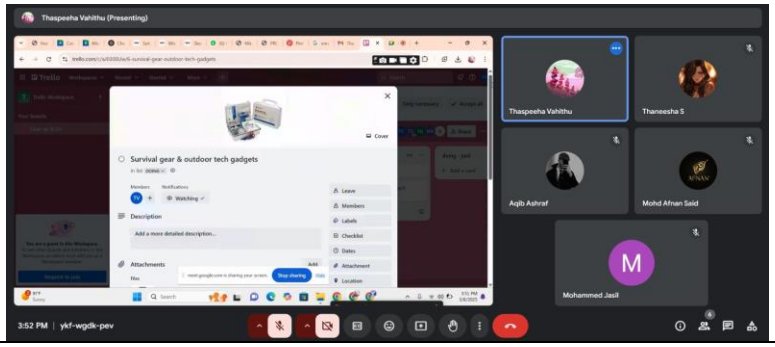
Stage 4	COLLABRATIVE DESIGN

49 | Page

Mode	Online via google meet
Status of attendence	All present
Date	March 10 – March 15
Minutes	1. Established a Trello board for task management. 2. Created a Gantt chart to track the 8-week timeline. 3. Constructed website pages for efficient navigation. 4. Created a Work Breakdown Structure (WBS) for task clarity. 5. Integrated features: filters, search bar, wishlist, and customer reviews.
Challenges	Image sizes caused delays in page loading.
Solutions	Optimized images to enhance performance.
Action items completd	Updated Trello; conducted design tests across devices.



(Google Meeting Link shared through WhatsApp)

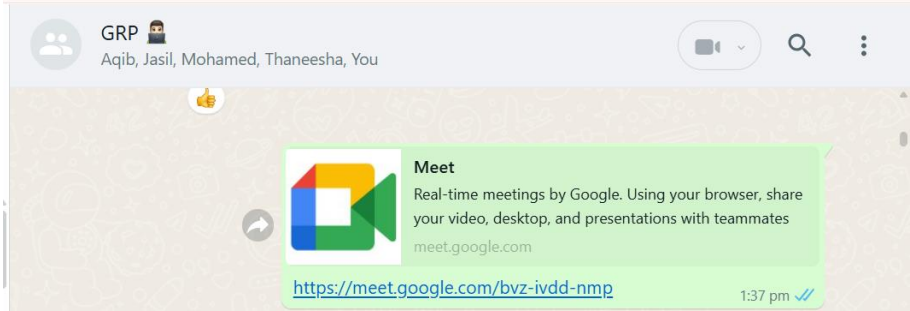


15/03/25 - 23/03/25	Offline Meeting	

Stage 5	**FEATURE IMPLEMENTATION AND TESTING**
Mode	In person at univeristy
Status of attendence	All present
Date	March 15 – march 23
Minutes	1. Tested inter-page linking and identified issues: slow loads and scroll failures. 2. Fixed errors by optimizing images and debugging scroll settings. 3. Finalized an Entity-Relationship diagram for product page databases.
Challenges	Testing revealed wrong search suggestions across pages
Solutions	Updated search bar to ensure uniform performance.
Action Items Completed	Documented resolutions; validated features.
Stage 6	**QUALITY ASSURANCE**
Mode	Online via google meet
Status of attendence	All present

21/03/25 - 31/03/25	Online Meeting	

Stage 6	**QUALITY ASSURANCE**
Mode	Online via google meet
Status of attendence	All present
Date	March 21 – March 31
Minutes	1. Reviewed website pages for functionality and consistency. 2. Standardized fonts and colors to align with branding. 3. Applied animations to improve user engagement. 4. Confirmed responsiveness across mobile, tablet, and desktop platforms.
Challenges	Mobile layouts exhibited inconsistencies.
Solutions	Redesigned responsive templates to maintain consistency.
Action Items Completed	Secured design; planned final testing.



(Google Meeting Link shared through whatsapp)

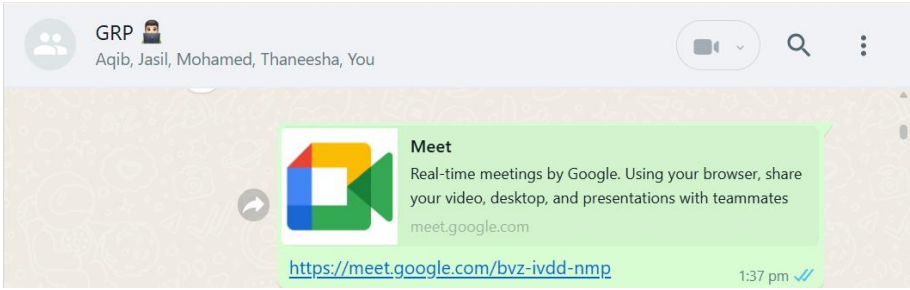
01/04/25 - 06/04/25	Offline Meeting	

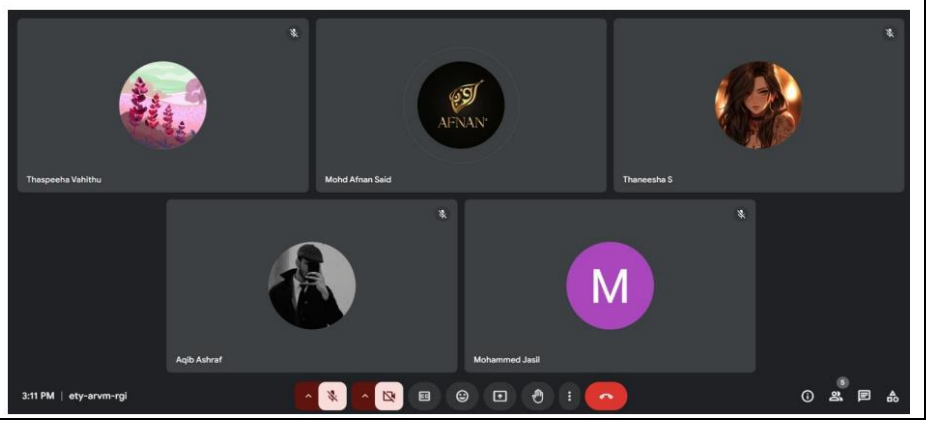
Stage 7	**FINAL INTEGRATION**
Mode	In person at univeristy
Status of attendence	All present
Date	April 1 – April 6
Minutes	1. Completed end-to-end testing, ensuring all features functioned.

50 | Page

[redacted]

	2. Installed a secure payment gateway for transactions. 3. Documented a team charter defining shared goals. 4. Compiled the final report in five sections: Introduction, Methodology, Development, Challenges, Conclusion.
Challenges	Managing testing, payment setup, and report compilation simultaneously.
Solutions	Tasks were divided efficiently, and regular check-ins kept the team on track.
Action items completed	Validated payments; finalized report drafts.

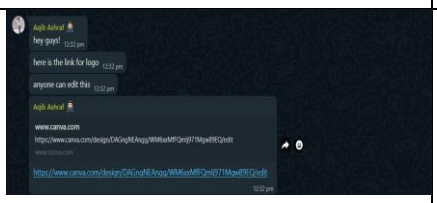
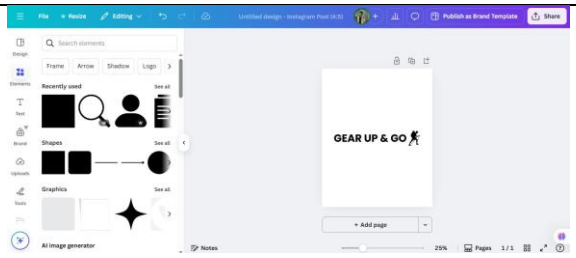
<table>
<tr>
<td>07/04/25 - 13/04/25</td>
<td>Online Meeting</td>
<td>
<table>
<tr><td>Stage 8</td><td>COMPLETION</td></tr>
<tr><td>Mode</td><td>Online via google meet</td></tr>
<tr><td>Status of attendence</td><td>All present</td></tr>
<tr><td>Date</td><td>April 7 – April 13</td></tr>
<tr><td>Minutes</td><td>Reviewed and proofread the final report.
Tested the website link across devices for responsiveness.
Prepared a PowerPoint presentation</td></tr>
<tr><td>Challenges</td><td>Noticed minor report typos</td></tr>
<tr><td>Solutions</td><td>Conducted a team review</td></tr>
<tr><td>Action items completed</td><td>Readied ppt for presentation , Finalized report for submission</td></tr>
</table>


(Google Meeting Link shared through Whatsapp)


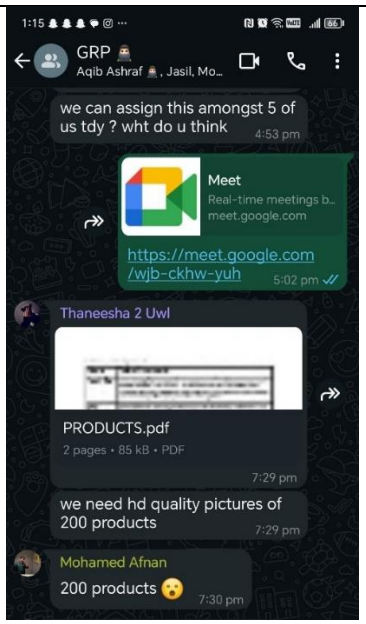
</td>
</tr>
</table>

Table 1-Details about minutes of meetings.

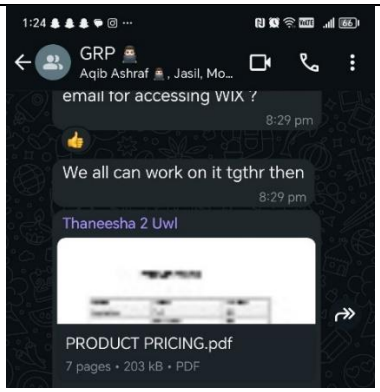
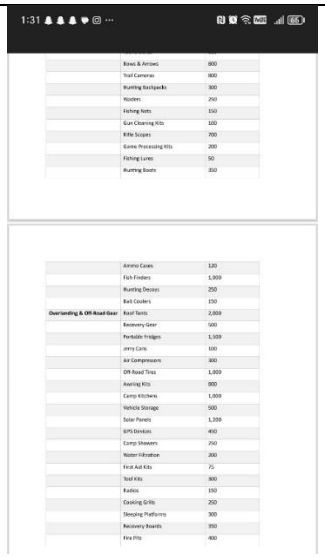
Evidence of shared documents

Date	Discussion of shared documents/evidence (WhatsApp)	Documents shared
16/02/2025 **Suggested and finalized topic.**		
1/3/25		

<table>
<tr>
<td>26/02/25

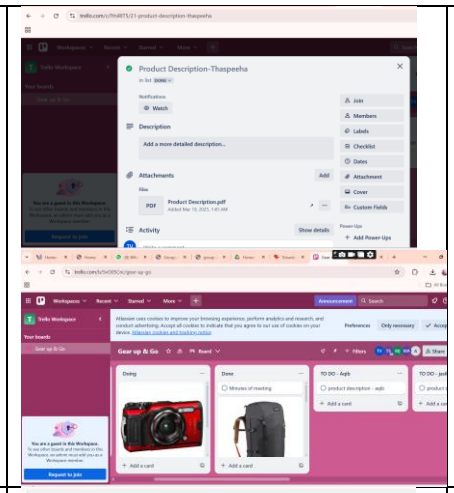
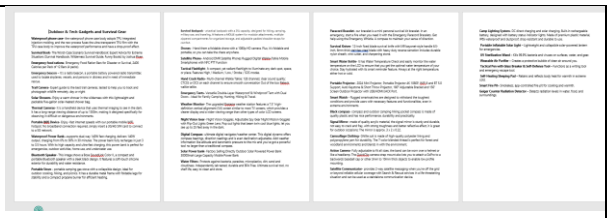
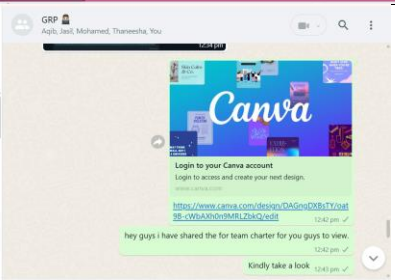
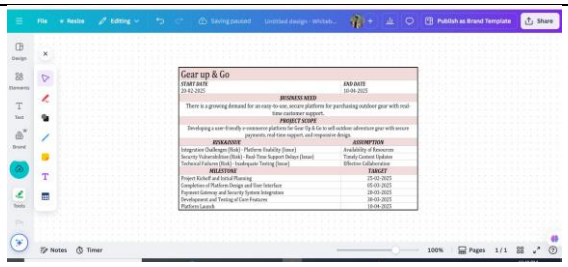
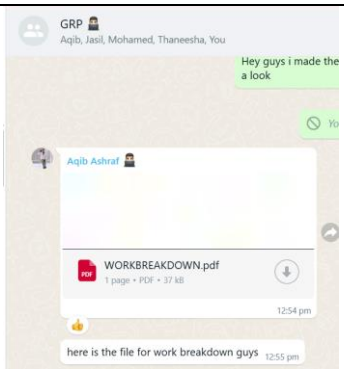
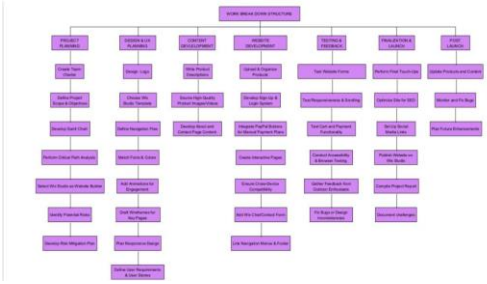
Gear Up & Go Products shared and finalized.</td>
<td> 
</td>
<td>
PRODUCTS (Total 200 products)

Category	Products (20 per category)
Camping Gear	Tents, Sleeping Bags, Camping Chairs, Camping Stoves, Portable Coolers, Multi-Tools, Camping Lanterns, First Aid Kits, Fire Starters, Water Purifiers, Cooking Sets, Inflatable Mattresses, Hammocks, Tent Repair Kits, Solar Chargers, Camping Tables, Waterproof Tarps, Bear Spray, Tent Fans, Camping Pillows
Hiking Equipment	Hiking Backpacks, Trekking Poles, Hydration Packs, GPS Devices, Headlamps, Hiking Boots, Gaiters, Trail Maps, Energy Bars, Hiking Pants, Rain Jackets, Compass, Emergency Blankets, Sunglasses, Portable Seats, Dry Bags, Hiking Hats, Lightweight Jackets, Sunscreen, Bug Spray
Outdoor Clothing	Base Layers, Insulated Jackets, Softshell Pants, Rainwear Sets, Quick-dry Shirts, Fleece Jackets, Hiking Socks, Gloves, Beanies, Neck Gaiters, Sun Hats, Waterproof Boots, Trail Running Shoes, Climbing Pants, Swimwear, Thermal Underwear, Convertible Pants, Windbreakers, Sandals, Overalls
Water Sports Gear	Kayaks, Paddleboards, Life Jackets, Waterproof Phone Cases, Dry Bags, Snorkel Sets, Fishing Rods, Water Shoes, Swim Goggles, Underwater Cameras, Inflatable Boats, Paddles, Wet Suits, Diving Knives, Tow Ropes, Buoyancy Belts, Floating Coolers, Spearfishing Gear, Waterproof Backpacks, Anchor Kits
Climbing Gear	Climbing Helmets, Harnesses, Carabiners, Ropes, Belay Devices, Climbing Shoes, Crash Pads, Chalk Bags, Climbing Gloves, Rope Bags, Ascenders, Quickdraws, Climbing Tape, Slings & Webbing, Portable Hangboards, Belt Kits, Climbing Pants, Pulley Systems, Climbing Backpacks, Rope Protectors
Cycling Gear	Mountain Bikes, Road Bikes, Helmets, Cycling Shoes, Bike Lights, Repair Kits, Hydration Packs, Panniers & Bags, Cycling Gloves, Jerseys, Bike Locks, GPS Computers, Multi-tools, Bike Racks, Knee & Elbow Pads, Tire Pumps, Chain Lubes, Bike Covers, Sunglasses, First Aid Kits
Survival Gear	Survival Knives, Paracord Bracelets, Emergency Tents, Solar Chargers, Tactical Flashlights, Water Filters, Signal Mirrors, Emergency Food Rations, Fire Starters, Multi-tools, Compass, First Aid Kits, Survival Backpacks, Bear Sprays, Satellite Phones, Emergency Blankets, Hand Crank Radios, Camouflage Clothing, Survival Books, Portable Stoves
Outdoor Tech & Gadgets	Action Cameras, Drones, Solar Power Banks, Handheld GPS Devices, Satellite Communicators, Weather Monitors, Bluetooth Speakers, Smartwatches, Portable Projectors, Thermal Cameras, Trail Cameras, Waterproof Power Banks, Camp Lighting Systems, Digital Compasses, Night Vision Gear, Solar Showers, Portable WiFi Devices, Emergency Beacons, Smart Water Bottles, Weatherproof Phone Cases
Hunting & Fishing Gear	Hunting Knives, Fishing Rods, Camouflage Clothing, Binoculars, Game Calls, Tackle Boxes, Bows & Arrows, Trail Cameras, Hunting Backpacks, Waders, Fishing Nets, Gun Cleaning Kits, Rifle Scopes, Game Processing Kits, Fishing Lures, Hunting Boots, Ammo Cases, Fish Finders, Hunting Decoys, Bait Coolers
Overlanding & Off-Road Gear	Roof Tents, Recovery Gear, Portable Fridges, Jerry Cans, Air Compressors, Off-Road Tires, Awning Kits, Camp Kitchens, Vehicle Storage, Solar Panels, GPS Devices, Camp Showers, Water Filtration, First Aid Kits, Tool Kits, Radios, Cooking Grills, Sleeping Platforms, Recovery Boards, Fire Pits
</td>
</tr>
<tr>
<td>7/03/25

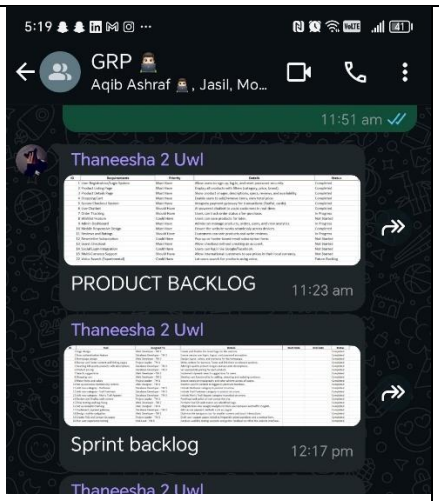
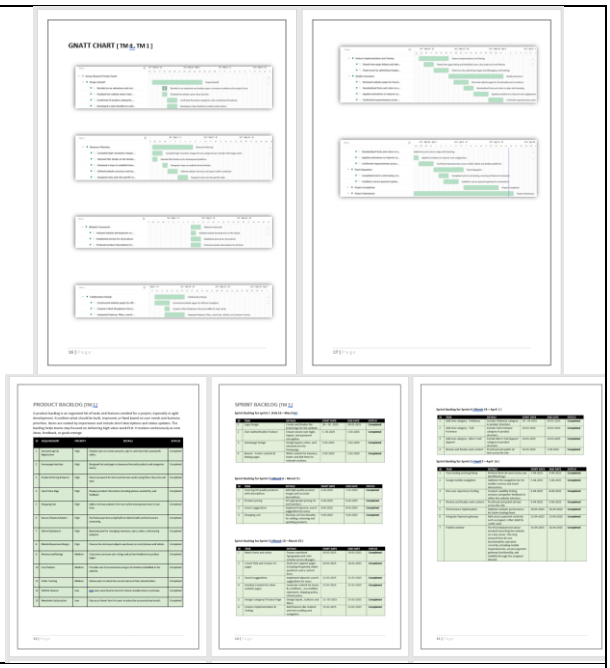
Product pricing shared and finalized</td>
<td> 
</td>
<td> 
1:31

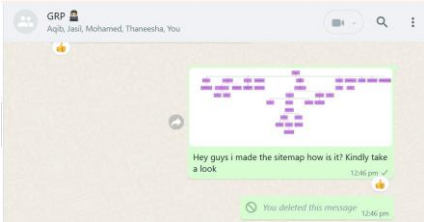
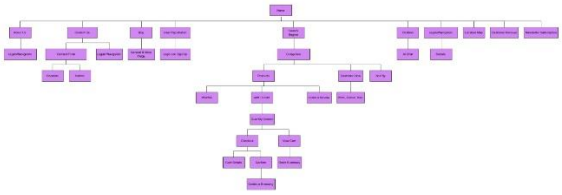
Bows & Arrows	800
Trail Cameras	800
Hunting Backpacks	300
Waders	250
Fishing Nets	150
Gun Cleaning Kits	100
Rifle Scopes	700
Game Processing Kits	200
Fishing Lures	50
Hunting Boots	350

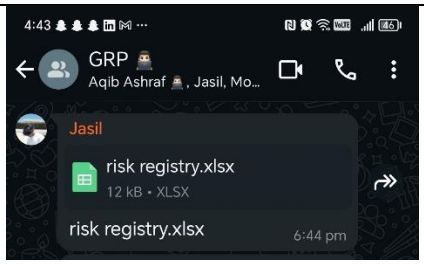
Ammo Cases	120
Fish Finders	1,000
Hunting Decoys	250
Bait Coolers	150
Overlanding & Off-Road Gear	Roof Tents 2,000
Recovery Gear	500
Portable Fridges	1,500
Jerry Cans	100
Air Compressors	300
Off-Road Tires	1,000
Awning Kits	800
Camp Kitchens	1,000
Vehicle Storage	500
Solar Panels	1,200
GPS Devices	450
Camp Showers	250
Water Filtration	200
First Aid Kits	75
Tool Kits	300
Radios	150
Cooking Grills	250
Sleeping Platforms	300
Recovery Boards	350
Fire Pits	400
</td>
</tr>
</table>

26/02/25-13/04/25 **Trello for task management**		
03/04/25 **Team Charter**		
13/03/25 **Work Breakdown Structure Shared and Finalized**		

<table>
<tr>
<td>11/03/25

Gantt Chart ,Sprint and Product Backlogs</td>
<td> 
</td>
<td>  </td>
</tr>
<tr>
<td>7/04/25

Sitemap shared and finalized.</td>
<td> 
</td>
<td>  </td>
</tr>
<tr>
<td>13/04/25

Risk Register shared.</td>
<td> 
</td>
<td>
<table>
<tr><th>ID</th><th>Risk Description</th><th>Likelihood</th><th>Impact</th><th>Severity</th><th>Owner</th><th>Mitigating Action</th><th>Contingent Action</th><th>Progress</th><th>Status</th></tr>
<tr><td>1</td><td>Design Misinterpretation</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>Use wireframes and design approvals</td><td>Revise based on feedback</td><td>Initial review done</td><td>OPEN</td></tr>
<tr><td>2</td><td>Technical Website Issues</td><td>High</td><td>High</td><td>High</td><td>Project Manager</td><td>Use reliable CMS and test continuously</td><td>Roll back to stable version</td><td>QA phase ongoing</td><td>OPEN</td></tr>
<tr><td>3</td><td>Payment Gateway Failure</td><td>Medium</td><td>High</td><td>High</td><td>Project Team</td><td>Test APIs (PayPal, Stripe) thoroughly</td><td>Use backup method</td><td>Testing done</td><td>OPEN</td></tr>
<tr><td>4</td><td>Scope Creep</td><td>High</td><td>High</td><td>High</td><td>Project Team</td><td>Lock scope with change log</td><td>Reassess timeline/resources</td><td>Scope finalized</td><td>OPEN</td></tr>
<tr><td>5</td><td>Data Privacy Breach</td><td>Low</td><td>High</td><td>High</td><td>Project Team</td><td>Follow GDPR, secure hosting</td><td>Notify users, patch system</td><td>Security review started</td><td>OPEN</td></tr>
<tr><td>6</td><td>Chatbot Malfunction</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>AI testing and fallback setup</td><td>Disable and report issue</td><td>Testing in progress</td><td>OPEN</td></tr>
<tr><td>7</td><td>Broken Links/404 Errors</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>Conduct regular audits</td><td>Set up 404 redirect</td><td>Audit ongoing</td><td>OPEN</td></tr>
<tr><td>8</td><td>Communication Gaps</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>Weekly meetings, task briefs</td><td>Assign team lead</td><td>Ongoing</td><td>OPEN</td></tr>
<tr><td>9</td><td>Budget Overrun</td><td>Medium</td><td>High</td><td>High</td><td>Project Team</td><td>Monitor resource usage</td><td>Adjust feature scope</td><td>Monitoring</td><td>OPEN</td></tr>
<tr><td>10</td><td>Incompatibility on Devices/Browsers</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>Perform cross-device testing</td><td>Fix layout/styles for compatibility</td><td>Testing phase</td><td>OPEN</td></tr>
<tr><td>11</td><td>Low User Engagement Post-Launch</td><td>Medium</td><td>Medium</td><td>Medium</td><td>Project Team</td><td>Add analytics, engaging content</td><td>Run promotions, gather feedback</td><td>Engagement plan drafted</td><td>OPEN</td></tr>
<tr><td>12</td><td>Failure in Backup or Recovery Plan</td><td>Low</td><td>High</td><td>High</td><td>Project Team</td><td>Automate backups, cloud sync</td><td>Manually restore from last backup</td><td>Backup plan reviewed</td><td>OPEN</td></tr>
</table>
</td>
</tr>
</table>

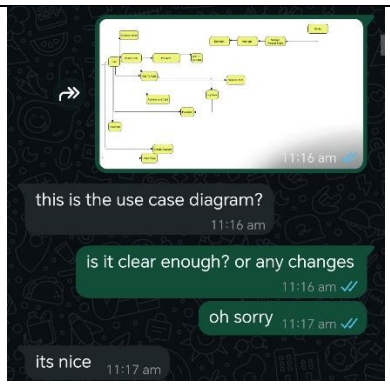
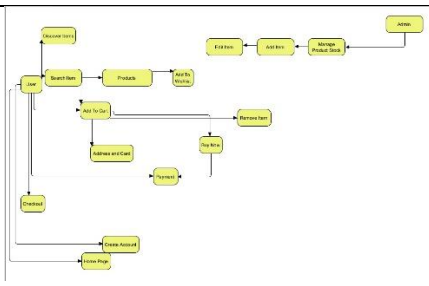
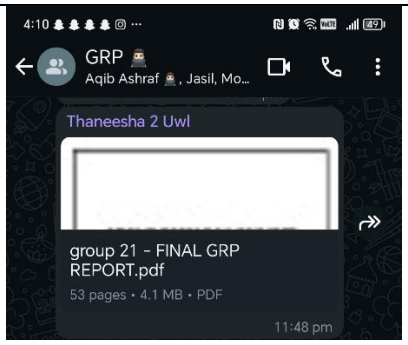
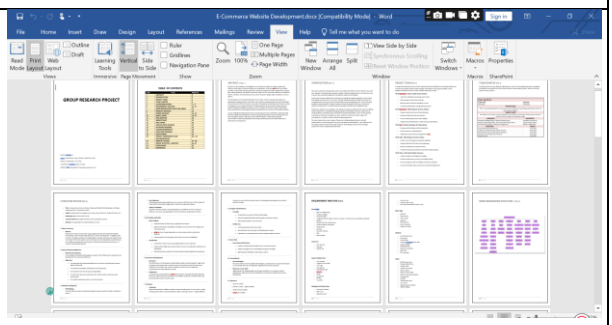
21/03/25 **User Case Diagram**		
13/04/25 **Final Group Report**		

Table 2-Table with details about documents shared.

Evidence of the contribution of each group member to the final project deliverables

TM.1] Thaneesha – Project Leader & Scrum Master

- Leads team meetings and manages sprint cycles.
- Tracks progress and resolves obstacles.
- Allocates tasks and ensures deadlines are met.
- Coordinates deliverables, including the final report.

TM.2] Thaspeeha – Web Designer & Systems Architect

- Designs website structure and visual elements.
- Oversees architecture and user interface.
- Ensures website performance and scalability.
- Collaborates with developers to meet technical requirements.

TM.3] Jasil – Database Developer & Technical Lead

- Manages website design and data integration.
- Ensures secure user authentication.
- Implements structured data management in Wix.

TM.4] Aqib – UML Designer & Systems Analyst

- Creates Use Case Diagrams and system workflows.
- Develops WBS and PERT Chart for task planning.
- Analyzes and optimizes system processes.
- Provides technical documentation and system design mapping.

TM.5] Afnan – Risk Lead & Quality Assurance

- Leads risk analysis and mitigation strategies.
- Conducts performance, security, and usability testing.
- Ensures compliance with accessibility and security standards.
- Documents testing results and resolves issues before deployment

Fig 1. Team Roles and Responsibilities

3. Challenges Faced

Firstly, we constantly browsed through topics, changed from Skincare to the pet store, and into choosing the right topic, which is our 'Gear Up & Go' focused on adventure and camping gear, to gear up our users for adventures.

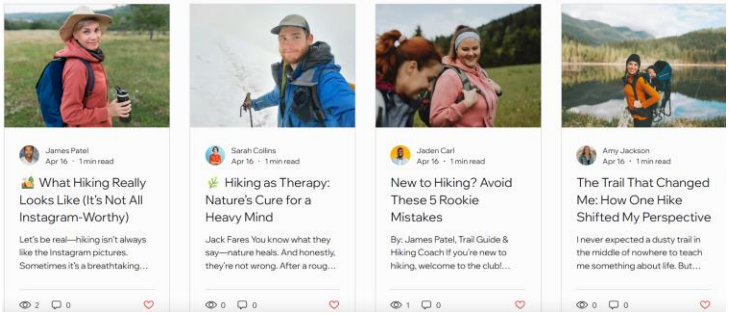
Secondly, our challenge came in choosing to either build the backend from scratch or use a web builder tool. Due to time constraints, we solved the issue by going with Wix. Creating a seamless, intuitive, and responsive design that works across all devices, especially mobile devices, was difficult due to a large catalog of products with extra designs/layouts.

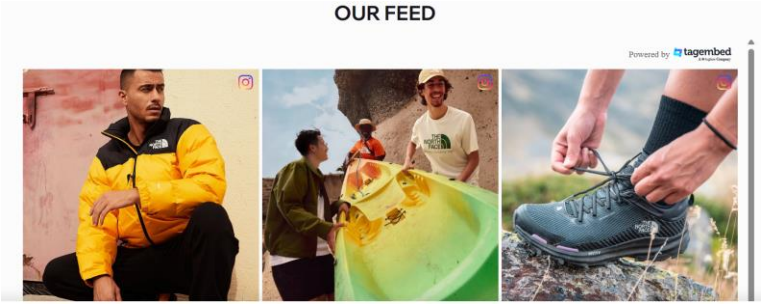
Supporting multiple payment gateways securely and reliably across regions, but it was not implemented due to limits in our Wix plan. Language Selector enhances usability across regions and for users, but was not implemented as well due to limits in our Wix plan.

Reflecting on the project, we identified areas for improvement. Using OneDrive and Outlook email for document sharing, instead of WhatsApp, would have enhanced accessibility. On a positive note, choosing visual paradigms as our design tool was a successful decision, as it met our needs effectively, and Trello was used for keeping our database of our project.

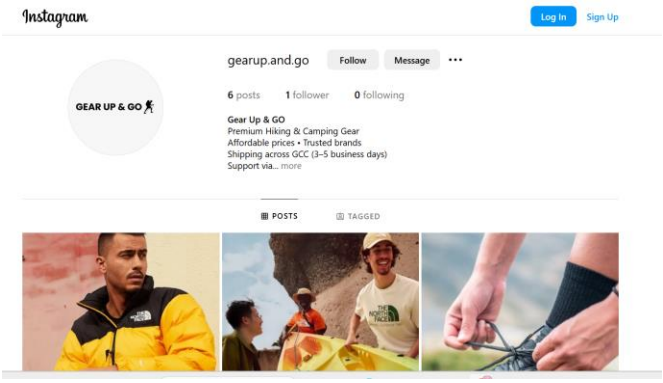
4. Innovation and Attention to Detail

1. Blog: Encourages community engagement and sharing of experiences related to adventure and outdoor lifestyles.



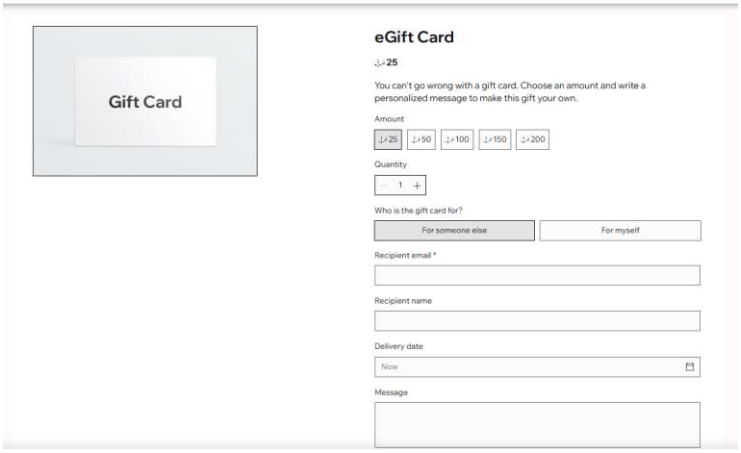


2. Socials Page: Users/customers can view our feed through our website. It includes customer engagement and products exposed through social media.

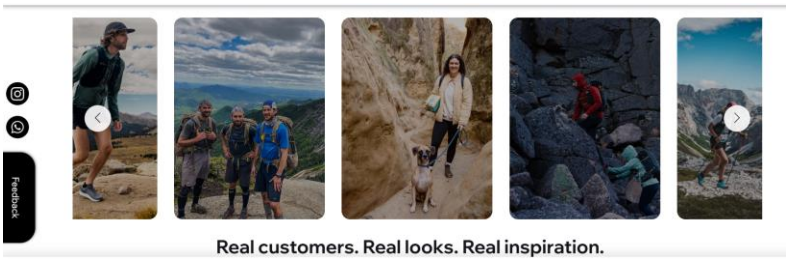


3. Feedback Form: Allows customers to provide valuable insights and help improve the overall shopping experience.

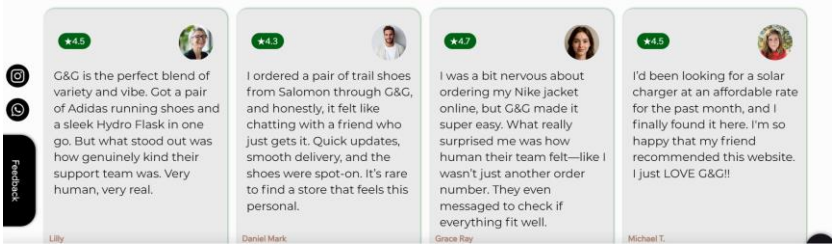
4. E-Gift Card: Allows users to buy their favourite explorer or for themselves to buy a digital gift card for 25,50,100,200 dhs with a personalized message.



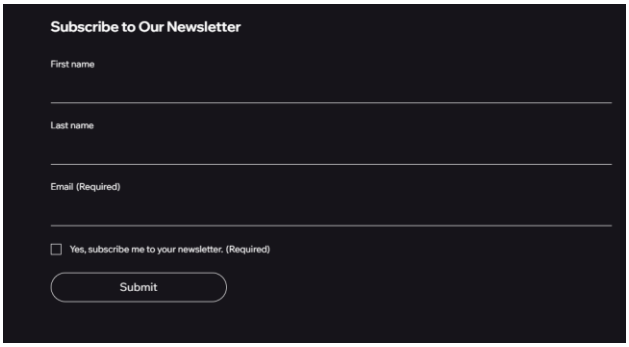
5. Sliding Gallery: Product pictures and adventurous or explored moments gallery.



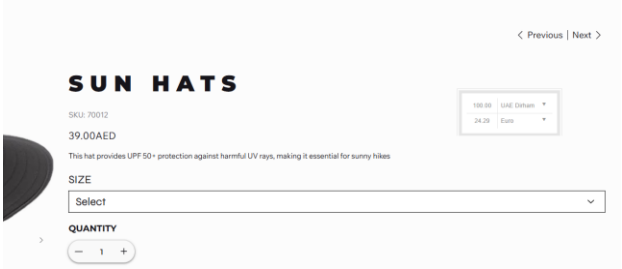
6. Customer Reviews: Provides social proof to enhance user confidence in the products.



7. Newsletter: Allows users to complete the sign-up process and join the mailing list to get updates.



8. Currency Converter: Easier to view the product cost in different currencies.



Task 2: (Research Project Proposal)

Contents

1. Introduction 19

2. Aims, Objectives, and Research Questions 19

2.1 Aim 19

2.2 Objectives 19

2.3 Research Questions 19

3. Literature Review 20

3.1 AI in Healthcare Decision Support 20

3.2 Neural Networks as Learning Agents 20

3.3 The Rise of Explainable AI (XAI) 20

3.4 Human-Centered Trust in AI 20

3.5 Ethical and Empowerment Dimensions 20

4. Proposed Research Project 21

4.1 Justification, Usefulness, and Feasibility 21

5. Methodology 22

5.1 Data and Tools 22

5.2 Model Development 22

5.3 User Interaction Simulation 22

6. Project Plan 22

7. Expected Outcomes 23

8. References 23

1. Introduction

Artificial Intelligence (AI) is transforming industries worldwide, with healthcare standing at the forefront. One of the most promising applications of AI is in **Healthcare Decision Support Systems (HDSS)**, where AI models assist in diagnostics, treatment planning, and risk prediction. However, these systems are often built on complex black-box models—like deep neural networks—whose decision-making logic is difficult to understand. This opacity poses a barrier to **trust, adoption, and usability**, especially in life-critical domains like medicine.

This proposal outlines a research project to explore how **Explainable AI (XAI)** methods can enhance trust in HDSS, ensuring AI decisions are transparent, understandable, and inclusive. As a woman in tech passionate about responsible AI, I believe explainability is essential not only for technical success but also for ethical empowerment—particularly in ensuring that AI uplifts, not undermines, human agency, including women's roles in healthcare.

2. Aims, Objectives, and Research Questions

2.1 Aim

To investigate how Explainable AI techniques can improve trust, usability, and inclusivity in AI-powered Healthcare Decision Support Systems.

2.2 Objectives

- To explore the state-of-the-art in XAI methods for healthcare.
- To build a neural network-based HDSS model.
- To integrate SHAP and LIME for model interpretability.
- To simulate user interaction and evaluate trust and usability through surveys.
- To examine how explainability intersects with ethical empowerment, especially for women in clinical roles.

2.3 Research Questions

1. Which XAI techniques most effectively improve clinician trust in HDSS?
2. How does explanation quality influence system usability and decision-making speed?
3. Can trust be quantified through user-centered metrics in a simulated clinical environment?
4. How can explainability support ethical and inclusive AI, particularly with respect to women's roles in healthcare?

3. Literature Review

3.1 AI in Healthcare Decision Support

AI-driven decision support tools have shown great potential in medical diagnostics (e.g., skin cancer detection using CNNs – Esteva et al., 2017), patient monitoring (Komorowski et al., 2018), and predictive modeling. Despite their accuracy, these tools are often mistrusted by healthcare professionals due to their opaque “black-box” nature (Amann et al., 2020).

3.2 Neural Networks as Learning Agents

Neural networks are a core component of AI agents—especially **learning agents** that interact with and adapt to human behavior. They mimic human cognitive processes by learning from patterns in data. However, their complexity often makes them difficult to interpret, which raises challenges in sensitive applications like healthcare where accountability is non-negotiable (Rudin, 2019).

3.3 The Rise of Explainable AI (XAI)

Explainable AI aims to bridge this gap by making model predictions more interpretable:

- **SHAP (SHapley Additive exPlanations)** attributes importance to features.
- **LIME** explains individual predictions using interpretable surrogate models.
- **Saliency maps** visually highlight regions influencing decisions in medical images.
- **Attention mechanisms** in neural nets show which inputs the model "attended" to most.

However, there is a trade-off between model accuracy and interpretability, a topic heavily debated in the field (Samek et al., 2019).

3.4 Human-Centered Trust in AI

Trust is not a purely technical metric; it’s a **human-centered attribute**. Research shows clinicians are more likely to adopt AI tools that offer clear, human-like explanations and allow for collaborative decision-making (Tonekaboni et al., 2019). The **usability** of explanations—whether doctors can understand and act upon them—is just as critical as the explanations themselves.

3.5 Ethical and Empowerment Dimensions

Explainability is not only about usability—it also intersects with **ethics and empowerment**. Without transparency, AI can reinforce existing biases or exclude marginalized groups, including women in healthcare decision-making roles. As a strong advocate for women's empowerment, I believe AI should serve as a tool for **inclusive innovation**, not a gatekeeper of authority.

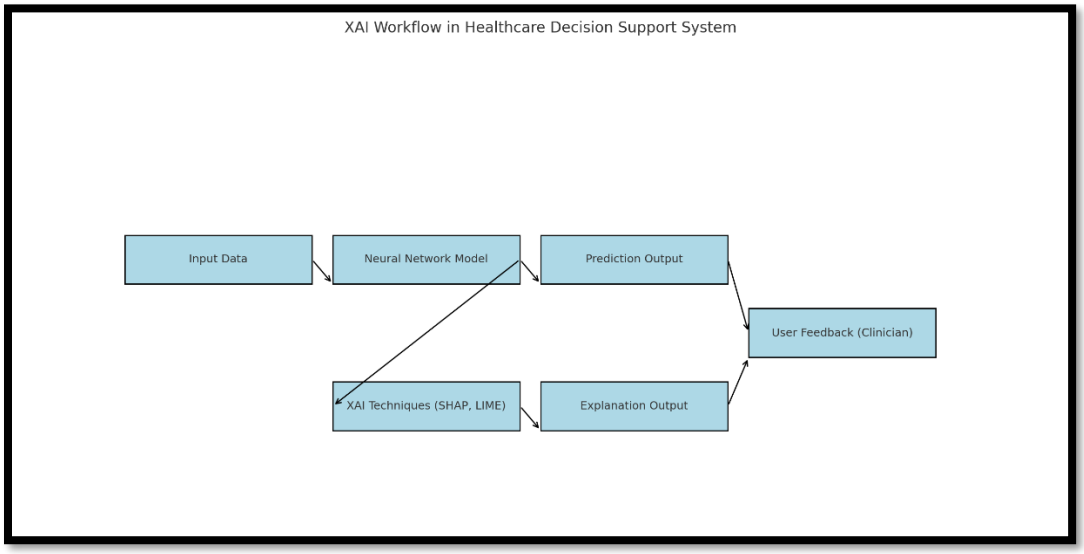


Fig 1. Adapted from concepts discussed in Tjoa and Guan (2020) and Holzinger et al. (2019), illustrates the workflow of integrating XAI into neural network-based healthcare decision support systems.

4. Proposed Research Project

4.1 Justification, Usefulness, and Feasibility

This research is:

- **Useful**: It addresses a core barrier to AI adoption in healthcare—lack of trust.
- **Feasible**: Tools like SHAP, LIME, and public datasets (e.g., MIMIC-III) make development and evaluation accessible.
- **Innovative**: It combines technical modeling with human-centered evaluation and ethical awareness.

The proposed work contributes both practically (by enhancing HDSS usability) and ethically (by promoting trustworthy and inclusive AI design).

5. Methodology

5.1 Data and Tools

- Dataset: MIMIC-III or Breast Cancer Diagnostic Dataset.
- Tools: Python, TensorFlow, SHAP, LIME, Captum (for model explanations).
- Evaluation: Surveys and trust-rating interfaces designed for medical professionals.

5.2 Model Development

A neural network will train a binary classification model (e.g., cancer vs. no cancer). SHAP and LIME will be applied to explain predictions.

5.3 User Interaction Simulation

Participants (ideally medical students or simulated users) will be shown AI predictions with and without explanations. A Likert scale and qualitative input will collect their trust, clarity rating, and feedback.

6.Project Plan

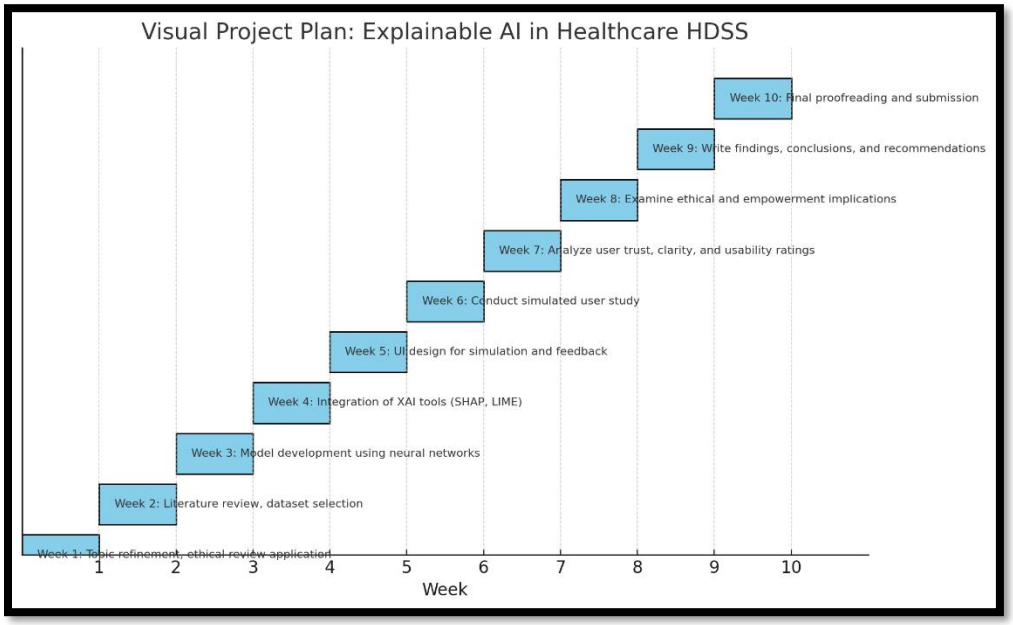


Fig 2. Gantt chart outlining the 10-week project plan for research on Explainable AI in Healthcare (Thaspeeha Vahithu, 2025).

7. Expected Outcomes

- A neural network-based HDSS with integrated explanations.
- Comparative analysis of SHAP and LIME effectiveness in trust-building.
- Insights into how explainability affects usability and decision confidence.
- Ethical reflection on how explainable AI can support inclusivity and women's empowerment in health tech.

8. References

1. Esteva, A., Kuprel, B., Novoa, R. A., et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature.
2. Komorowski, M., et al. (2018). The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis. Nature Medicine.
3. Doshi-Velez, F., & Kim, B. (2017). Towards A Rigorous Science of Interpretable Machine Learning. arXiv:1702.08608.
4. Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions. Nature Machine Intelligence.
5. Samek, W., Montavon, G., et al. (2019). Explainable AI: Interpreting, Explaining and Visualizing Deep Learning. Springer.
6. Tonekaboni, S., et al. (2019). What Clinicians Want: Contextualizing Explainable ML for Clinical End Use. ML for Healthcare.
7. Amann, J., et al. (2020). Explainability for Artificial Intelligence in Healthcare: A Multidisciplinary Perspective. BMC Medical Informatics.
8. Tjoa, E. and Guan, C., 2020. A survey on explainable artificial intelligence (XAI): Toward medical XAI. IEEE Transactions on Neural Networks and Learning Systems, 32(11), pp.4793–4813.
9. Holzinger, A., Langs, G., Denk, H., Zatloukal, K. and Müller, H., 2019. Causability and explainability of artificial intelligence in medicine. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 9(4), p.e1312.
10. Lundberg, S.M. and Lee, S.I., 2017. A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems, 30.
11. Thaspeeha Vahithu, 2025. *Gantt chart of 10-week project plan for Explainable AI in Healthcare research*. [Fig 2].